THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 12 workers can finish a job in 20 days. 16 workers will finish it in: 1

(a) 17 days **(b)** 15 days

(c) 26.666666666666666666666667 days **(d)** 16 days

2. A car covers 180 km in 3 hours at a steady speed. In 5 hours it covers: 1

(a) 300 km **(b)** 0.083333333333333333333333333333 km

(c) 185 km **(d)** 360 km

3. ₹ 1800 is divided among A, B and C in the ratio 2 : 3 : 4. The smallest share is: 1

(a) ₹ 600 **(b)** ₹ 400 **(c)** ₹ 800 **(d)** ₹ 360

4. The cost of articles and the number of articles (same price each) are in: 1

(a) no proportion **(b)** equal ratio always 1 : 1

(c) inverse proportion **(d)** direct proportion

5. The ratio of 3 m 20 cm to 80 cm is: 1

(a) 4 : 1 **(b)** 1 : 4 **(c)** 5 : 1 **(d)** 4 : 2

6. If A : B = 1 : 2 and B : C = 6 : 5, then A : B : C is: 1

(a) 3 : 6 : 5 **(b)** 1 : 6 : 5 **(c)** 1 : 2 : 5 **(d)** 1 : 2 : 30

7. On a map, 1 cm represents 8 km. A distance of 6.5 cm on the map is actually: 1

(a) 60 km **(b)** 52 km **(c)** 5.2 km **(d)** 14.5 km

8. 6 workers can finish a job in 24 days. 9 workers will finish it in: 1

(a) 18 days **(b)** 16 days **(c)** 36 days **(d)** 21 days

9. A car covers 330 km in 5 hours at a steady speed. In 8 hours it covers: 1

(a) 594 km **(b)** 338 km

(c) 528 km **(d)** 0.121212121212121212121212121212 km

10. ₹ 2400 is divided among A, B and C in the ratio 3 : 4 : 5. The smallest share is: 1

(a) ₹ 800 **(b)** ₹ 600 **(c)** ₹ 480 **(d)** ₹ 1000

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. 14 workers complete a task in 15 days. How many days will 10 workers take? 2

12. A map scale is 1 cm = 30 km. Find the actual distance for 4.2 cm and the map distance for 150 km. 2

13. Find the ratio of 5 m 40 cm to 180 cm in simplest form. 2

14. 20 workers complete a task in 9 days. How many days will 15 workers take? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If A : B = 1 : 2 and B : C = 6 : 5, find A : B : C. 3

16. 8 men can build a wall in 30 days. How many days will 12 men take to build it? 3

17. The ages of two friends are in the ratio 4 : 7 and the sum of their ages is 33 years. Find their ages after 6 years. 3

OR

If A : B = 4 : 5 and B : C = 10 : 9, find A : B : C. 3

18. 14 men can build a wall in 21 days. How many days will 7 men take to build it? 3

OR

The ages of two friends are in the ratio 5 : 6 and the sum of their ages is 44 years. Find their ages after 8 years. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Divide ₹ 2400 among A, B and C in the ratio 3 : 4 : 5. If C gives ₹ 300 to A, find the new ratio of their amounts. 5

OR

A bus covers 330 km in 5 hours at a uniform speed. Find 5
(i) its speed
(ii) the distance it covers in 8 hours
(iii) the time it takes to cover 462 km.

20. Divide ₹ 1500 among A, B and C in the ratio 2 : 5 : 8. If C gives ₹ 100 to A, find the new ratio of their amounts. 5

OR

A bus covers 150 km in 2 hours at a uniform speed. Find 5
(i) its speed
(ii) the distance it covers in 9 hours
(iii) the time it takes to cover 600 km.

END OF QUESTION PAPER — Total: 40 Marks

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